Instruction for the use of Perl scripts

Basic setup

all scripts depend on subroutines stored in the perl module Biomodule.pm.

place the Biomodule.pm and all WICKERsoft scripts into a dedicated directory which is in a common path (e.g. /usr/local/bin/) so that you can run them from anywhere in your system.

make sure that all scripts are executable. if necessary, make them executable with chmod 755 <script_name>

change the path in the beginning of all WICKERsoft scripts to that directory example, how the header should look:

```
#!/usr/bin/perl
use lib '/usr/local/bin/';
use Biomodule;
```

Additional software needed:

From standard Ubuntu repositories:

Emboss
Clustalw
blastall / ncbi-blast+-legacy (local stand-alone blast program)
perl-tk
dotter
libgd-graph-perl
libgd-perl
libgd-text-perl

Blast programs
the Perl scripts described here use the **blastall** program from ubuntu repositories. Searchable databases were produced with the format **formatdb** that come along with the package.
By default they are placed in a directory /home/wicker/data/databases/. This path has to be adjusted in the scripts.
Blast requires the BLOSUM62 matrix which is also provided in the repository.

All original Perl scripts used here show a **usage manual** when the program name is typed without arguments (this shows a list of the required arguments)

Procedures for this study are listed below

If there are questions or problems, contact wicker@botinst.uzh.ch

Chromosome colienarity analysis

This protocol is used to identify colinear chromosomal region, TE presence/absence polymorphisms (PAPs) and deletions

In this study this procedure for the chromosome conservation plots and identifications of deletions and TE insertions in Supplementary Figure S18.

Alignments of megabase-sized chromosomal fragments chromosomes are aligned with the Perl script **blast_compare_chromosome**. This runs blastn searches of a given chromosome from species/haplotype against the homologous chromosome of the other species/haplotype. It splits the sequence 1 in user-defined (e.g. 1 kb) windows with a user-defined sliding step between windows (e.g. 500 kb for glabel, 1 kb for local alignments) and uses them in blastn searches against the other chromosome.
The blast output is parsed an the position of the best hit on sequence 2 is recorded. Furthermore, the top alignment is extracted and the numbers of SNPs is recorded. Transitions and transversions are treated separately in case they are used for molecular dating (this was not used in the study).
The program produces a log file with the name prefix **"log_chr_comp_"**

The header line has names and sizes of chromosomes (or chromosome segments) used

The columns in the log file are as follows:
1. ID of segment compared
2. start position of segment used for blastn on Bgt_96224_v316_chr11
3. start position of top hit on Bgs_1451_chr11
4. number of aligned bases
5. Sequence identity
6. Number of transitions (A ↔ G and C ↔ T)
7. Number of transverions (all other substitutions)
8. Divergence time estimate based on a substitution rate of 1.3E-8
9. Standard deviation of divergence time estimate

Chromosome dot plots
The log files produced by blast_compare_chromosome are visualized as dot plots with the Perl script **visual_chromosome_compare_dotplot**. This uses the Perl-Tk module for an interactive GUI. It produces a simple dot plot matrix from the positions of the best blastn hits. For the plots in the figure, the option "color_hits" and color_contrast can be used for the adjustment of the color sacle for sequence identity.

Molecular dating
the divergence times of species/haplotypes are estimated based to the chromosome blastn alignments produced with **blast_compare_chromosome**. For molecular dating, the average over multiple blast hits (e.g. 100) is then used (see below for details).

Procedures used in this study are described in more detail below

Comparison of centromeric and peri-centromeric regions

Step 1: Identification of large colinear segments
Large colinear segments are identified in a first overall comparison in which the centromeric regions (centromeres plus ~50 Mb of flanking regions) were compared with **blast_compare_chromosome** using a segment size of 1000 and a sliding step of 500 kb.
Log files were visualized with **visual_chromosome_compare_dotplot**.

Identification of TE presence/absence polymorphisms (PAPs) was subsequently done with selected chromosomal segments of 5-10 Mb which show good colinearity so that they can be aligned reliably (Fig.1).

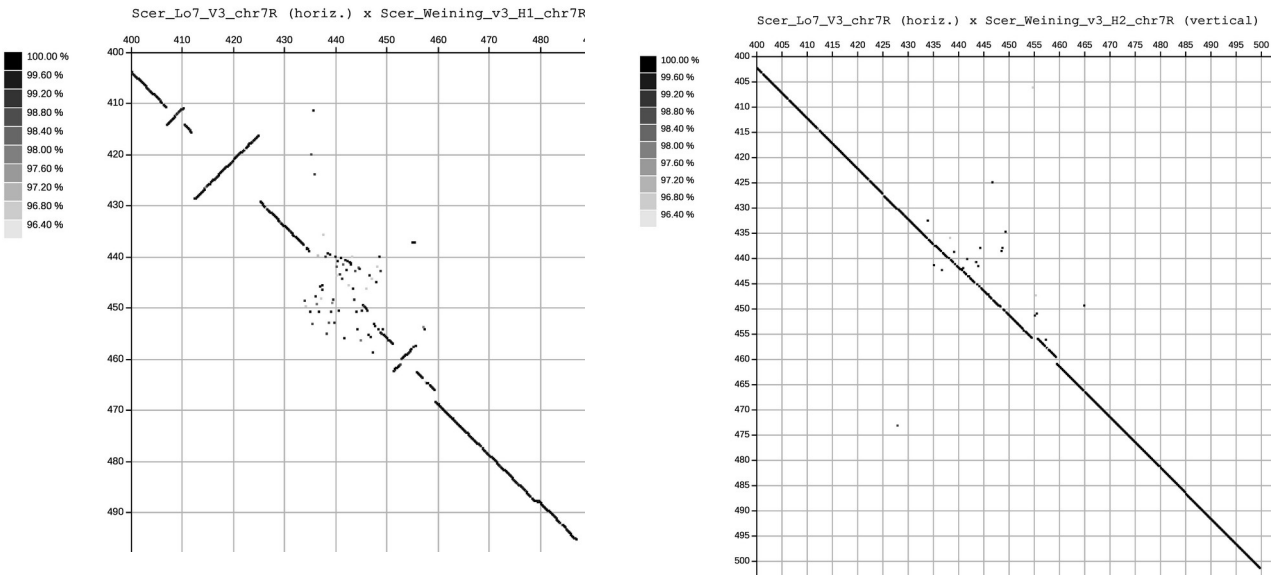

Fig 1. The example plots (produced with **visual_chromosome_compare_dotplot**) show comparison of chromosome 7R from rye Lo7 against Weining haplotype 1 and 2 in the region 400-500 Mb. Note that H2 shows much lower sequence colinearity with Lo7 than H1. Consequently, fewer segments could be select for later local alignment.

Step 2: Based on the first global alignment colinear segments (in this study including the centromere were defined. Selected colinear segments are excised from the chromosomes of both species with the script **assemble**. A total of 6 regions were selected, rangig in size from ~5 to 15 Mb:

```
# comparison Lo7 x H2 ++++++++++++++++++++++++++++++++++++++++++++
Segments                            Position [Mb]
Scer_Lo7_V3_chr7R_centL             425-435
Scer_Weining_v3_H2_chr7R_centL      427-437

Scer_Lo7_V3_chr7R_cent              435-445
Scer_Weining_v3_H2_chr7R_cent       437-447

Scer_Lo7_V3_chr7R_centR             445-455
Scer_Weining_v3_H2_chr7R_centR      447-457

Scer_Lo7_V3_chr7R_colin3            460-475
Scer_Weining_v3_H1_chr7R_colin3     463-478

# comparison Lo7 x H1 ++++++++++++++++++++++++++++++++++++++++++++
Segments                            Position [Mb]
Scer_Lo7_V3_chr7R_inv               451.1-455.8
Scer_Weining_v3_H1_chr7R_inv        457.0-462.4 reverse

Scer_Lo7_V3_chr7R_colin             460-475
Scer_Weining_v3_H1_chr7R_colin      468-484
```

Positions (in Mb) of the selected segments are also provided in **positions_colin_segments**.

Step 3: The excised colinear fragments from the two species were aligned again with **blast_compare_chromosome** using a segment size of 1000 and a sliding step of 1000 bp (in order to generate quasi gap-free alignments subsequently). This was done in both directions, as the following steps only identifies insertions in one direction (e.g. insertions in species 2, when species 1 is aligned with species 2).

the 12 resulting log_chr_comp_ files are provided as zip archives:
log_chr_comp_rye_Lo7_x_H1_x_H2.tar.gz
log_chr_comp_rye_H1_x_H2_Lo7.tar.gz

Step 4: the program **TE_PAP_from_log_chr_comp** was used to identify large (i.e. TE-sized) insertions in species 2. The program first scans the log_file of **blast_compare_chromosome** for colinear regions using the following criteria:

1. All blast hits in reverse orientation are dismissed
2. Non-colinear hits are identified by calculating the median over 30 position of hits in species 2. All hits that deviate from the median by more than 35 kb are dismissed as hits to non-colinear loci.
3. the remaining filtered blast hits are then screened for insertion in species 2. that are between 3kb and and 17 kb.c
4. The program runs a dotplot alignments using the software dotter (unutu repositories) for visual inspection of the candidate InDel regions and asks the used whether the InDel candidate should be kept
5. If the region is deemed a true indel, the program then runs a local alignment using the Smith-Waterman algorithm implemented in the program water (EMBOSS package) and excises the largest insertion in species 2.

Step 5. The isolated insertion sequences were used as queries against the nrTREP25 TE database to identify homologies with know TE families

Step6. The program **png_blast_coverage_TE_all** is used to manually examine the blast outputs to verify insertions of full-length TEs and distiguish them from deletions of random segments.
- The program produces a graphical representaion of blast hits in linear and dot plot format This allows easy visual inspection if an InDel corresponds to an insertion of a full-length element (i.e. the full TE consensus aligns with the InDel).

Step 7.
Annotations were then visualized with **visual_chromosome_compare_segment**. A screenshot of the display was used for Fig S20a.

The script uses as inputs the a log_chr_comp plus the associated annotation files, for example:

log_chr_comp_Scer_Lo7_V3_chr7R_cent__x__Scer_Weining_v3_H2_chr7R_cent.tab
annotation_insertions_Scer_Lo7_V3_chr7R_cent
annotation_insertions_Scer_Weining_v3_H2_chr7R_cent

the annotation files for the segments are provided in
annotations_chr_segments.tar.gz

Step 8. molecular dating of haplotype divergence times.
The program **date_from_log_chr_comp** uses the same algorithm to identify colinear segments as **TE_PAP_from_log_chr_comp** described above.

- It uses the **log_chr_comp_** ouputs from the local alignments of the selected segments.
- It calculates divergence time in user-defined sliding windows (e.g. 400 windows, sliding step of 50) along the compared selected regions
- It adds up total aligned bp and numbers of transitions and transversions for each window and used them to estimate divergence time.
- It use the substitution rate of 1.5E-8 per site per year that is routinely used for molecular dating of regions that are free from selection pressure (i.e. gene-free TE regions: Buchmann JP bet al., 2012. Plant J. 71:550-63. DOI: 10.1111/j.1365-313X.2012.05007.x). Since the regions analysed here are practically void of genes, it was assumed that they are complete gene-free

For this analysis, the comparison Lo7 vs. H1 and H2 were used.
The data is provided in **log_chr_comp_rye_Lo7_x_H1_x_H2.tar.gz**

The output file **chr_comp_date_log_all_win400** was used as input for the panel Fig S20c

Step 9. The InDels that represented insertions of full-length TEs were used for molecular dating of insertion ages (i.e. through comparisons of LTRs)

the resulting insertion age table chr_comp_TE_pos_age_log_all was used for the boxplot shown in Fig 20d and the graphical linear annotation plot in Fig 20e

The R-scripts used to produce the plots in Fig 20c through S20e are provided in the file **Fig_S20_colinearity_species_divergence_TE_PAP.R**